ACCESS POINTS

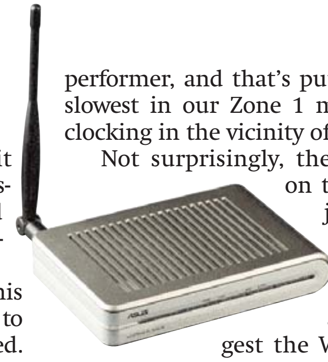
Five Access Points (APs) competed in our test. An AP is the simplest way to set up a Wi-Fi network at home. It's convenient to connect a single laptop to a PC. SoHos should look elsewhere, because APs typically have just a single RJ45 connect.

ASUS WL-320G

Cool looker, light performer

The ASUS WL-320G is wall-mountable; since it doesn't have rubber foot pegs it will move easily if kept on a surface. It's well built, though, and the activity LEDs are well-placed and clearly legible.

ASUS has incorporated a high-speed mode on this model—Afterburner. The client device also needs to support it, in which case performance is boosted. Fancy titles aside, the WL-320G was a very mediocre performer, and that's putting it mildly. It was the slowest in our Zone 1 multiple file transfer test, clocking in the vicinity of 400 seconds.



Not surprisingly, the lower data throughput on the WL-320G led to some jerks during our movie streaming test. Although it wasn't downright unwatchable, we wouldn't suggest the WL-320G for this specific purpose, especially at the steep price of Rs 8,900.

Compex WP54G

An AP not worth accessing!

The third in the Compex trio to be received was a simple AP—no routing capabilities here.

The WP54G supports PoE (Power over Ethernet). Although PoE implementations differ greatly between manufacturers, the basic principle remains the same (see box *Wi-Fi Demystified*).

The WP54G lagged behind in quite a few tests, especially in Zone 3, where it was thrashed by the competition. In fact, data transfer was interrupted once, due to a loss in signal integrity.

Video streaming, as we'd expected, was abysmal—jerking, and lots of interruptions. We don't recommend the WP54G as an AP, despite the inclusion of PoE—simply because of its inadequate performance. Its price tag of Rs 5,950 won't garner any accolades either.

Wi-Fi Demystified

Wi-Fi: An acronym for Wireless Fidelity, Wi-Fi quite simply denotes "Wireless Networks" that conform to IEEE (International Electrical and Electronics Engineers) standards. The Wireless standard accepted by the IEEE is 802.11. The most common Wi-FI standards within 802.11 are 802.11a, 802.11b, 802.11g, 802.11 Super g, and 802.11 Pre n.

Gateway: A gateway is responsible for proper data flow and distribution within a network and externally to other networks. The gateway machine is also called the host computer or network router.

SSID: Service Set Identifier or simply "Network Name" is the name (typically alphanumeric) by which a Wireless Local Area Network (WLAN) identifies itself. To communicate with any devices on a wireless network, all wireless devices must use the same SSID. The SSID can be kept private, in which case a user who doesn't know the SSID cannot gain access to that network.

WEP: Acronym for Wired Equivalent Privacy. This was the earliest security method adopted by the 802.11 standard, providing 64- and 128-bit keys (hexadecimal coded) to protect wireless devices from unwanted intrusion.

WPA/WPA2: An acronym for Wireless Protected Access, which is a more secure method of protecting data transmission than WEP. WPA 2 is a newer protocol. It's also known as 802.11i.

DHCP: Acronym for Dynamic Host Configuration Protocol. It's a communications protocol that automatically assigns IP addresses to clients logging on to a TCP/IP network.

MAC Address: An acronym for Media Access Control Address. It's a unique identifier for an individual node on a Wi-Fi network. It is represented as a string of six pairs of hex characters, for example, FF.EE.FF.FF.00.FF

NAT: An acronym for Network Address Translation, it's basically a firewall feature. NAT is available on most routers today, both wired and otherwise. NAT enables individual IPs of clients on a network LAN to remain invisible beyond the LAN, and any external communication occurs through the gateway machine. Packets of data recieved by the gateway machine are then routed to the clients that requested them. In case the packet received doesn't correspond with a request sent by a machine on a network, it is rejected.

Port Mapping: Port Mapping or Port Address Translation is the process of translating and routing packets arriving at a particular IP address or port to another IP address. Port Mapping bypasses NAT.

Deep Packet Inspection: Deep Packet Inspection (DPI) is a form of packet filtering that inspects the data portion of data packets as opposed to just checking the header portion. DPI is basically an evolved form of SPI (Stateful Packet Inspection).

PoE (Power over Ethernet): This describes the ability to transmit power as well as data to devices over an Ethernet cable. PoE is extremely useful in cases where the AP is placed in an area with no power point.